AI-CONTROLLED HIGH-PERFORMANCE UNCREWED AIRCRAFT WITH CONFIDENCE-GATED ADAPTIVE FLIGHT MANAGEMENT

Patent preparation package - U.S. utility/nonprovisional focus

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Important Use Note

This package is a patent drafting aid, not legal advice and not a representation that the claims are patentable. Inventor names, ownership, dates, priority claims, government-funding disclosures, and any public disclosure dates must be verified before filing. The technical concept is intentionally framed around aircraft autonomy, safety, health management, sensor fusion, and authenticated human supervision; it does not claim autonomous weapon targeting or engagement.

Proposed Core Concept

A high-performance uncrewed aircraft with an onboard autonomy stack that fuses heterogeneous sensor data into a confidence-scored state estimate, generates and continuously revises flight plans under explicit safety envelopes, uses aircraft-health predictions to reconfigure flight-control and mission behavior, and accepts human commands through cryptographically authenticated, capability-scoped mission contracts. When communications degrade or AI confidence falls below a configured threshold, the aircraft transitions through verified fallback behaviors and preserves an auditable mission-state record for reconciliation after link recovery.

AI-CONTROLLED HIGH-PERFORMANCE UNCREWED AIRCRAFT WITH CONFIDENCE-GATED ADAPTIVE FLIGHT MANAGEMENT

Draft specification - to be filed in DOCX for a U.S. nonprovisional utility application

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to [PROVISIONAL APPLICATION NUMBER/DATE], if applicable. If no provisional or other priority application is used, delete this section.

FIELD OF THE INVENTION

[0002] The present disclosure relates to uncrewed aircraft systems and, more particularly, to high-performance uncrewed aircraft having onboard artificial-intelligence-assisted flight management, multimodal sensor fusion, adaptive health-aware control, communications resilience, and cryptographically authenticated human supervision.

BACKGROUND

[0003] Uncrewed aircraft are increasingly capable of operating with limited direct control. Conventional architectures often separate sensing, navigation, mission planning, aircraft health monitoring, communications, and flight control into software layers with limited shared state. Such separation can create inconsistent assumptions when sensor confidence changes, components degrade, communications are interrupted, or a remote operator modifies a mission during flight.

[0004] Existing autonomous flight systems also may rely on a single mission plan generated before launch, or on planners that optimize a route without continuously coupling route selection to aircraft health, sensor confidence, communications quality, and verified operating constraints. A technical need exists for an integrated onboard architecture that continuously converts uncertain data into a controlled flight state while constraining AI-generated behaviors through deterministic safety logic and preserving verifiable mission history.

SUMMARY OF THE INVENTION

[0005] In one aspect, the invention provides an uncrewed aircraft comprising a flight-control computer, heterogeneous sensors, an onboard AI processor, a state-estimation engine, a mission-contract interpreter, a safety-envelope engine, a health-state estimator, a communications manager, and an auditable event log.

[0006] The state-estimation engine produces a fused aircraft/environment state together with confidence measures and provenance identifiers for source measurements. The AI processor generates candidate navigation or task-management actions. The safety-envelope engine evaluates candidate actions against aircraft limits, configured operating constraints, current health state, and confidence thresholds before an action is made available to the flight controller.

[0007] Human-originated instructions are represented as cryptographically authenticated mission contracts containing permitted action classes, temporal validity, geographic or operational constraints, and revocation information. The aircraft verifies the contract and executes only actions within the authenticated authority.

[0008] When communications quality degrades, the communications manager selects a configured degraded-link policy and stores unresolved mission-state differences. Upon link recovery, a reconciliation engine merges state using timestamps, monotonic sequence numbers, provenance, and authority rules, thereby reducing stale or conflicting command application.

[0009] In another aspect, predicted component health is incorporated into planning before the aircraft commits to a new trajectory. A failure-prediction model can estimate a remaining-useful-operation interval or confidence of an impending degradation event, and the planner can choose among safe alternatives while preserving a bounded operating envelope.

[0010] These features may be implemented independently or in combinations, and can improve autonomy reliability, auditability, communications resilience, and controlled adaptation in high-performance uncrewed aircraft.

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 illustrates an example high-performance uncrewed aircraft and onboard computing architecture.

[0012] FIG. 2 illustrates the onboard autonomy stack and data paths.

[0013] FIG. 3 illustrates multimodal sensor fusion and confidence-aware state estimation.

[0014] FIG. 4 illustrates AI planning, candidate-action generation, and deterministic safety-envelope gating.

[0015] FIG. 5 illustrates health prediction and health-aware flight reconfiguration.

[0016] FIG. 6 illustrates authenticated human supervision and resilient communications.

[0017] FIG. 7 illustrates mission-state management and degraded-link reconciliation.

[0018] FIG. 8 illustrates an onboard compute/data architecture and evidence logging flow.

DETAILED DESCRIPTION

Architecture and airframe

[0019] FIG. 1 depicts an uncrewed aircraft 100 having a high-performance airframe 102, propulsion system 104, flight-control surfaces 106, power system 108, communications subsystem 110, sensor suite 112, onboard compute subsystem 114, and data-storage subsystem 116. The airframe may be fixed-wing and may have a configuration broadly resembling a contemporary fighter-type aircraft, although the inventive features are not limited to any particular aerodynamic layout. The propulsion system may include one or more engines or electric propulsion units. The aircraft may optionally include non-weapon mission payload interfaces for sensing, navigation, communications relay, or other lawful non-kinetic functions.

Onboard autonomy stack

[0020] FIG. 2 shows software and hardware components including a sensor-ingest layer 200, feature extraction layer 202, state-estimation engine 204, environment model 206, mission-contract interpreter 208, AI planner 210, safety-envelope engine 212, health-state estimator 214, flight-control interface 216, communications manager 218, evidence logger 220, and reconciliation engine 222. Components can execute on one processor, multiple processors, graphics processing units, field-programmable gate arrays, safety-certified controllers, or combinations thereof.

Sensor fusion

[0021] FIG. 3 illustrates fusion of inertial, air-data, navigation, optical, infrared, radar, lidar, acoustic, or other sensor measurements. A measurement may be tagged with source identity, acquisition time, calibration version, quality score, and cryptographic integrity status. The state-estimation engine can generate a fused state vector containing aircraft position, velocity, attitude, angular rates, energy state, and confidence intervals. Environment features can be represented separately from aircraft state and can carry uncertainty estimates.

Confidence and provenance

[0022] Confidence values are propagated from sensor measurements through state estimation and planning. Instead of treating an AI prediction as an unqualified command, the system can attach a

confidence score and provenance graph to each candidate action. Provenance identifies the model version, input-data set identifiers, sensor sources, and relevant state-estimation snapshot. A threshold policy may require additional verification, human confirmation, alternate sensing, or transition to a conservative mode when confidence is insufficient.

Mission contracts

[0023] A mission contract 230 is a machine-readable instruction object authenticated by a trusted authority. The contract may include mission identifier, issuer identifier, validity period, permitted action classes, operational constraints, geographic constraints, resource limits, revocation state, and cryptographic signature. The aircraft verifies the signature and authority chain before interpreting the instructions. A contract can be changed or revoked without replacing the aircraft's core flight software.

AI planning

[0024] The AI planner 210 generates one or more candidate trajectories or navigation behaviors from mission objectives and the current state. Candidate generation can use model-predictive control, learned policies, reinforcement learning, search, heuristic planning, or hybrid approaches. The invention does not require a particular learning method. The significant control relationship is that candidate outputs are not sent directly to actuators; they pass through the safety-envelope engine.

Safety envelope

[0025] The safety-envelope engine 212 enforces deterministic constraints associated with aircraft structural limits, propulsion limits, control authority, energy reserves, sensor validity, communications status, configured airspace restrictions, and other authorized operating constraints. The engine can reject a candidate, modify it to satisfy constraints, select another candidate, or command a verified fallback maneuver. The constraints can be parameterized by aircraft configuration and health state.

Health-aware adaptation

[0026] FIG. 5 shows a health-state estimator 214 receiving motor or engine telemetry, vibration measurements, electrical data, thermal measurements, actuator feedback, and control residuals. A machine-learning model can predict a fault class or degradation probability. The planner incorporates the health estimate into trajectory selection and resource allocation. For example, when a predicted degradation reduces expected maneuver authority, the system can select a trajectory that maintains a larger margin to configured limits and increases recovery options.

Redundant control and mode transition

[0027] A verified mode manager can maintain multiple control configurations, including nominal autonomy, degraded autonomy, remote-supervision-assisted control, conservative navigation, and safe recovery. Transition logic can use hysteresis and persistence conditions to avoid mode oscillation. A safety controller with independent computation can override AI outputs when a protected constraint is violated.

Communications resilience

[0028] FIG. 6 shows a communications manager 218 that monitors link quality, latency, packet loss, authentication state, and channel availability. The manager can select among available links or relay paths. When link quality falls below a threshold, the aircraft can enter a degraded-link state in which only previously authorized action classes remain executable. The system records pending operator changes rather than treating a partially received message as executable.

Reconciliation after link recovery

[0029] FIG. 7 shows the reconciliation engine 222 maintaining a local mission-state record. State entries may include monotonic sequence numbers, timestamps, authority level, provenance, and conflict status. When communication is restored, the engine compares local and remote records and applies a deterministic reconciliation policy. A stale command may be rejected if superseded, an authoritative revocation may take precedence over a local plan, and unresolved conflicts can be surfaced for human review.

Simulation and shadow evaluation

[0030] In some implementations, a candidate AI action is first evaluated in a lightweight forward model that predicts aircraft response over a short horizon. The predicted result is compared with safety thresholds and with an uncertainty budget. The same evaluator may operate in a shadow mode during flight to compare an alternative policy without commanding the aircraft. Data from the comparison may be recorded for later model validation.

Model lifecycle management

[0031] AI models can be referenced by version identifiers, signing keys, validation status, and approved deployment scope. A model update can be accepted only if a trusted policy verifies the model signature, compatibility metadata, validation status, and authorized deployment window. The aircraft may retain a rollback model to support controlled reversion if a deployed model becomes invalid or unreliable.

Audit evidence

[0032] The evidence logger 220 stores selected state transitions, model identifiers, confidence values, command provenance, safety-envelope decisions, health estimates, and communications events. The log can be append-only or tamper-evident using hash chaining or another integrity mechanism. The resulting evidence enables after-action reconstruction without requiring every raw sensor sample to be retained.

Distributed implementation

[0033] Some functions may be distributed among the aircraft, a companion aircraft, a ground station, or a remote compute system. The aircraft remains capable of enforcing the local safety envelope even when remote services are unavailable. A distributed deployment can use deterministic versioned interfaces so that differences in model versions or state-estimation versions can be detected and isolated.

Training and validation

[0034] Training data may include historical flight data, synthetic simulation data, controlled test flights, and fault-injection scenarios. Validation may measure state-estimation error, trajectory tracking, constraint-violation rates, recovery performance, latency, and model-confidence calibration. A deployment policy can prevent models from being activated when validation results fall outside configured bounds.

Alternative embodiments

[0035] The described functions can be implemented using other machine-learning architectures, deterministic estimation methods, different sensor combinations, or different communications protocols. The aircraft may be piloted remotely, partially autonomous, or highly autonomous for non-weaponized functions. The scope of the invention is defined by the claims rather than by any particular hardware vendor, software framework, neural-network architecture, or airframe shape.

Industrial applicability

[0036] The disclosed architecture can be used in high-performance uncrewed aircraft for training, testing, reconnaissance, search and rescue, communications relay, environmental monitoring, flight research, logistics, or other lawful missions requiring reliable autonomous flight and controlled human supervision.

REFERENCE NUMERALS

100 aircraft; 102 airframe; 104 propulsion system; 106 flight-control surfaces; 108 power system; 110 communications subsystem; 112 sensor suite; 114 onboard compute subsystem; 116 storage subsystem; 200 sensor ingest; 202 feature extraction; 204 state estimation; 206 environment model; 208 mission-contract interpreter; 210 AI planner; 212 safety-envelope engine; 214 health-state estimator; 216 flight-control interface; 218 communications manager; 220 evidence logger; 222 reconciliation engine; 230 mission contract; 240 human supervisory station.

CLAIMS

1. An uncrewed aircraft comprising: an airframe; a plurality of heterogeneous sensors; an onboard processor; a state-estimation engine configured to generate an aircraft state and associated confidence data from outputs of the plurality of heterogeneous sensors; an artificial-intelligence planner configured to generate one or more candidate flight-control or navigation actions based on the aircraft state; a safety-envelope engine configured to evaluate the candidate actions against deterministic aircraft operating constraints and the confidence data; and a flight-control interface configured to provide to an aircraft controller only an action that is accepted or modified by the safety-envelope engine.

2. The uncrewed aircraft of claim 1, wherein the confidence data includes provenance identifiers linking a component of the aircraft state to one or more source sensors, source timestamps, and a model or estimator version.

3. The uncrewed aircraft of claim 1, further comprising a health-state estimator configured to determine a predicted degradation condition of at least one aircraft subsystem, wherein the safety-envelope engine modifies one or more operating constraints according to the predicted degradation condition.

4. The uncrewed aircraft of claim 3, wherein the artificial-intelligence planner generates at least a first candidate trajectory and a second candidate trajectory and selects between the candidate trajectories using the predicted degradation condition and a remaining operating margin.

5. The uncrewed aircraft of claim 1, further comprising a mission-contract interpreter configured to receive a cryptographically authenticated machine-readable mission contract specifying an authorized action set, validity interval, and one or more operating constraints, and to expose to the artificial-intelligence planner only instructions permitted by the mission contract.

6. The uncrewed aircraft of claim 5, wherein the mission-contract interpreter verifies a digital signature and rejects an instruction received outside the validity interval or outside an authorized action class.

7. The uncrewed aircraft of claim 1, further comprising a communications manager configured to detect degradation of a communication link and transition the aircraft to a degraded-link operating mode in which a restricted set of previously authenticated actions remains executable.

8. The uncrewed aircraft of claim 7, further comprising a reconciliation engine configured to store a local mission-state record while the communication link is degraded and reconcile the local

mission-state record with a remote mission-state record after communication recovery using sequence identifiers, timestamps, authority information, and provenance information.

9. The uncrewed aircraft of claim 1, wherein the safety-envelope engine includes an independent safety controller separate from the artificial-intelligence planner and configured to prevent an action that would violate a protected aircraft limit.

10. The uncrewed aircraft of claim 1, wherein at least one candidate action is evaluated using a forward simulation of predicted aircraft response before being provided to the flight-control interface.

11. A computer-implemented method for controlling an uncrewed aircraft, the method comprising: receiving heterogeneous sensor data; generating a fused aircraft state with confidence values and provenance; generating, by an artificial-intelligence planner, candidate navigation actions; evaluating the candidate navigation actions against a deterministic safety envelope; incorporating a predicted health state of an aircraft subsystem into the evaluation; and commanding an aircraft controller using a candidate navigation action that satisfies the deterministic safety envelope.

12. The method of claim 11, further comprising receiving a signed mission contract from a human supervisory system, verifying the mission contract, and constraining the candidate navigation actions to an authorized action set specified by the mission contract.

13. The method of claim 11, further comprising detecting communication degradation, preserving a local mission-state record, and entering a degraded-link policy that excludes unauthenticated or newly received actions until communication integrity is restored.

14. The method of claim 13, further comprising, after communication recovery, comparing the local mission-state record with a remote mission-state record and rejecting a remote command identified as stale according to sequence information and authority information.

15. The method of claim 11, further comprising recording model version information, confidence values, safety-envelope decisions, aircraft health state, and command provenance in a tamper-evident event log.

16. The method of claim 15, further comprising using the tamper-evident event log to reconstruct a sequence of AI-generated candidate actions and corresponding safety-envelope acceptance or rejection decisions.

17. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of an uncrewed aircraft, cause the processors to: generate a fused aircraft state from multiple sensors; generate candidate navigation actions using a machine-learning model; assign confidence data to the candidate navigation actions; evaluate the candidate navigation

actions using deterministic safety constraints and a predicted aircraft health state; and output a command to a flight controller only when the candidate navigation action satisfies the deterministic safety constraints.

18. The non-transitory computer-readable medium of claim 17, further comprising instructions for verifying a signed model package before activating the machine-learning model, the signed model package including a model identifier, compatibility information, validation status, and authorized deployment scope.

19. The non-transitory computer-readable medium of claim 17, further comprising instructions for selecting a fallback flight mode when confidence data falls below a threshold or when the communications manager detects loss of authenticated command connectivity.

20. A high-performance uncrewed aircraft system comprising the uncrewed aircraft of claim 1 and a human supervisory station configured to issue cryptographically authenticated mission contracts and receive evidence records comprising state estimates, AI model identifiers, confidence data, safety-envelope decisions, and communications-state transitions.

ABSTRACT OF THE DISCLOSURE

An uncrewed aircraft includes an onboard artificial-intelligence planner, heterogeneous sensors, a confidence-aware state-estimation engine, a deterministic safety-envelope engine, a health-state estimator, a cryptographically authenticated mission-contract interpreter, a resilient communications manager, and an auditable event logger. Sensor measurements are fused into an aircraft and environment state with confidence and provenance information. The AI planner generates candidate navigation actions, which are evaluated against aircraft limits, operating constraints, sensor confidence, communications status, and predicted subsystem health before reaching a flight controller. Human instructions are accepted through signed, capability-scoped mission contracts. During communications degradation, previously authenticated behaviors are maintained while unresolved state changes are recorded. After link recovery, local and remote mission states are reconciled using authority, sequence, timestamp, and provenance information. The architecture provides controlled adaptation and traceable operation for high-performance uncrewed aircraft.

CLAIMS AND ABSTRACT

Companion draft extracted from the specification for patent prosecution review

Draft Claims

1. An uncrewed aircraft comprising: an airframe; a plurality of heterogeneous sensors; an onboard processor; a state-estimation engine configured to generate an aircraft state and associated confidence data from outputs of the plurality of heterogeneous sensors; an artificial-intelligence planner configured to generate one or more candidate flight-control or navigation actions based on the aircraft state; a safety-envelope engine configured to evaluate the candidate actions against deterministic aircraft operating constraints and the confidence data; and a flight-control interface configured to provide to an aircraft controller only an action that is accepted or modified by the safety-envelope engine.

2. The uncrewed aircraft of claim 1, wherein the confidence data includes provenance identifiers linking a component of the aircraft state to one or more source sensors, source timestamps, and a model or estimator version.

3. The uncrewed aircraft of claim 1, further comprising a health-state estimator configured to determine a predicted degradation condition of at least one aircraft subsystem, wherein the safety-envelope engine modifies one or more operating constraints according to the predicted degradation condition.

4. The uncrewed aircraft of claim 3, wherein the artificial-intelligence planner generates at least a first candidate trajectory and a second candidate trajectory and selects between the candidate trajectories using the predicted degradation condition and a remaining operating margin.

5. The uncrewed aircraft of claim 1, further comprising a mission-contract interpreter configured to receive a cryptographically authenticated machine-readable mission contract specifying an authorized action set, validity interval, and one or more operating constraints, and to expose to the artificial-intelligence planner only instructions permitted by the mission contract.

6. The uncrewed aircraft of claim 5, wherein the mission-contract interpreter verifies a digital signature and rejects an instruction received outside the validity interval or outside an authorized action class.

7. The uncrewed aircraft of claim 1, further comprising a communications manager configured to detect degradation of a communication link and transition the aircraft to a degraded-link operating mode in which a restricted set of previously authenticated actions remains executable.

8. The uncrewed aircraft of claim 7, further comprising a reconciliation engine configured to store a local mission-state record while the communication link is degraded and reconcile the local mission-state record with a remote mission-state record after communication recovery using sequence identifiers, timestamps, authority information, and provenance information.

9. The uncrewed aircraft of claim 1, wherein the safety-envelope engine includes an independent safety controller separate from the artificial-intelligence planner and configured to prevent an action that would violate a protected aircraft limit.

10. The uncrewed aircraft of claim 1, wherein at least one candidate action is evaluated using a forward simulation of predicted aircraft response before being provided to the flight-control interface.

11. A computer-implemented method for controlling an uncrewed aircraft, the method comprising: receiving heterogeneous sensor data; generating a fused aircraft state with confidence values and provenance; generating, by an artificial-intelligence planner, candidate navigation actions; evaluating the candidate navigation actions against a deterministic safety envelope; incorporating a predicted health state of an aircraft subsystem into the evaluation; and commanding an aircraft controller using a candidate navigation action that satisfies the deterministic safety envelope.

12. The method of claim 11, further comprising receiving a signed mission contract from a human supervisory system, verifying the mission contract, and constraining the candidate navigation actions to an authorized action set specified by the mission contract.

13. The method of claim 11, further comprising detecting communication degradation, preserving a local mission-state record, and entering a degraded-link policy that excludes unauthenticated or newly received actions until communication integrity is restored.

14. The method of claim 13, further comprising, after communication recovery, comparing the local mission-state record with a remote mission-state record and rejecting a remote command identified as stale according to sequence information and authority information.

15. The method of claim 11, further comprising recording model version information, confidence values, safety-envelope decisions, aircraft health state, and command provenance in a tamper-evident event log.

16. The method of claim 15, further comprising using the tamper-evident event log to reconstruct a sequence of AI-generated candidate actions and corresponding safety-envelope acceptance or rejection decisions.

17. A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of an uncrewed aircraft, cause the processors to: generate a fused aircraft state from multiple sensors; generate candidate navigation actions using a machine-learning model; assign confidence data to the candidate navigation actions; evaluate the candidate navigation actions using deterministic safety constraints and a predicted aircraft health state; and output a command to a flight controller only when the candidate navigation action satisfies the deterministic safety constraints.

18. The non-transitory computer-readable medium of claim 17, further comprising instructions for verifying a signed model package before activating the machine-learning model, the signed model package including a model identifier, compatibility information, validation status, and authorized deployment scope.

19. The non-transitory computer-readable medium of claim 17, further comprising instructions for selecting a fallback flight mode when confidence data falls below a threshold or when the communications manager detects loss of authenticated command connectivity.

20. A high-performance uncrewed aircraft system comprising the uncrewed aircraft of claim 1 and a human supervisory station configured to issue cryptographically authenticated mission contracts and receive evidence records comprising state estimates, AI model identifiers, confidence data, safety-envelope decisions, and communications-state transitions.

Abstract

An uncrewed aircraft includes an onboard artificial-intelligence planner, heterogeneous sensors, a confidence-aware state-estimation engine, a deterministic safety-envelope engine, a health-state estimator, a cryptographically authenticated mission-contract interpreter, a resilient communications manager, and an auditable event logger. Sensor measurements are fused into an aircraft and environment state with confidence and provenance information. The AI planner generates candidate navigation actions, which are evaluated against aircraft limits, operating constraints, sensor confidence, communications status, and predicted subsystem health before reaching a flight controller. Human instructions are accepted through signed, capability-scoped mission contracts. During communications degradation, previously authenticated behaviors are maintained while unresolved state changes are recorded. After link recovery, local and remote mission states are reconciled using authority, sequence, timestamp, and provenance information. The architecture provides controlled adaptation and traceable operation for high-performance uncrewed aircraft.

APPLICATION DATA SHEET WORKSHEET (USPTO AIA/14 PREPARATION)

Do not file this worksheet as a substitute for the official USPTO ADS form.

Application Data

Field	Entry
Application type	☐ Nonprovisional utility ☐ Provisional ☐ Other: _____
Title	AI-Controlled High-Performance Uncrewed Aircraft with Confidence-Gated Adaptive Flight Management
First named inventor	[FULL LEGAL NAME]
Inventor citizenship	[COUNTRY]
Correspondence address	[ADDRESS]
Email / phone	[CONTACT]
Entity status	☐ Large ☐ Small ☐ Micro
Applicant / assignee	[LEGAL ENTITY OR INVENTOR]
Domestic benefit claim	[PROVISIONAL NO./DATE, IF ANY]
Foreign priority	[COUNTRY / APPLICATION NO. / DATE, IF ANY]
Government interest	☐ None ☐ Yes - identify agency/contract: _____
Attorney docket	[OPTIONAL]

Inventorship Verification

Identify every person who contributed to the conception of at least one claimed invention. Do not list a person solely because the person reduced the invention to practice, funded the work, supervised work, or built a conventional implementation unless that activity also involved conception of claimed subject matter.

Filing Note

The USPTO recommends routine use of an ADS for nonprovisional applications and requires an ADS for certain priority and benefit claims. The official filing should be prepared on the current USPTO form or Patent Center web-based ADS.

INVENTOR OATH / DECLARATION WORKSHEET

Use official USPTO PTO/AIA/01 or PTO/AIA/08 for filing; this worksheet is for preparation.

Inventor Information

Inventor	Legal name / residence / mailing address / citizenship
Inventor 1	[_____]
Inventor 2	[_____]
Inventor 3	[_____]
Inventor 4	[_____]

Declaration Checklist

- Confirm each listed inventor is an original inventor or original joint inventor of at least one claimed invention.
- Confirm the application was made or authorized to be made by each inventor.
- Confirm legal names and residence/mailing information are consistent with the ADS.
- Sign the official USPTO declaration in the permitted electronic or handwritten manner.
- If an inventor cannot sign, consult the current USPTO procedure for a substitute statement.

Suggested Identification Text

Application title: AI-Controlled High-Performance Uncrewed Aircraft with Confidence-Gated Adaptive Flight Management

Inventor(s): [INSERT EXACT LEGAL NAMES].

The official declaration should not be drafted from memory; use the current USPTO form language and confirm the current requirements before execution.

INFORMATION DISCLOSURE STATEMENT (IDS) DRAFT AND PRELIMINARY PRIOR-ART LIST

For attorney review and transfer to the official USPTO IDS form

Preliminary Search Scope

The preliminary search focused on autonomous UAV mission planning, multi-agent autonomy, sensor fusion, autonomous operation, and AI-based fault prediction. This list is intentionally preliminary; a professional patentability search should expand across U.S., PCT, EP, WIPO, and non-patent literature databases.

Publication	Topic / relevance
US20220351628A1	Automated mission planning and execution for a UAV; pairs aircraft with flight paths and transmits control instructions.
US7451023B2	Collaborative system for a team of unmanned vehicles; mission planning, situational awareness, and contingency management.
US12061673B1	Multi-agent planning and autonomy using learning-based approaches.
US20230280762A1	Autonomous vehicle mission planning and multi-asset collaborative operations.
US20200348696A1	Control of multi-drone fleets with temporal-logic objectives.
US10976752B2	Autonomous operation of UAVs; planning, coordination, and adaptation to uncertainty/faults.
US20250239091A1	Sensor fusion for autonomous machines using machine learning.
US20240210963A1	Failure prediction and risk mitigation in small uncrewed aircraft systems using AI/ML.

Potential Distinguishing Combination

- Confidence/provenance-aware sensor fusion tightly coupled to the planner.
- Deterministic safety-envelope gating of AI-generated flight actions.
- Health-prediction-driven reparameterization of the safety envelope and candidate trajectory selection.
- Cryptographically authenticated, capability-scoped mission contracts with explicit command authority.
- Degraded-link state preservation and deterministic reconciliation of local/remote mission state using authority + sequence + provenance.
- Tamper-evident evidence logging tying model version, confidence, health state, safety decisions, and command provenance together.

IDS Handling Note

An actual IDS should be prepared on the current USPTO form and should include copies or the bibliographic identifiers required under current rules. Do not assume this preliminary list is complete or that any cited document is legally material after claim amendments.

USPTO FILING AND FEE CHECKLIST

U.S. nonprovisional utility application

Core Filing Documents

- ☐ Application transmittal or equivalent Patent Center filing data
- ☐ Application Data Sheet (official PTO/AIA/14 / web ADS as applicable)
- ☐ Specification including description, claims, and abstract in DOCX
- ☐ Patent drawings, when necessary, in compliant black-and-white form
- ☐ Inventor oath or declaration using current USPTO form
- ☐ Filing fee
- ☐ Search fee
- ☐ Examination fee

Before Filing

- ☐ Confirm inventorship
- ☐ Confirm ownership / assignment chain
- ☐ Confirm any provisional priority claim and deadline
- ☐ Identify any foreign priority claims
- ☐ Identify any government funding or government-interest statement
- ☐ Confirm no unlisted public disclosure that may affect rights
- ☐ Review claim terminology for support in the specification
- ☐ Verify every claimed feature appears in the drawings when drawing support is needed

After Filing

- ☐ Save USPTO filing receipt
- ☐ Confirm application number and filing date
- ☐ Track restriction / office action deadlines
- ☐ File IDSs as appropriate
- ☐ Monitor maintenance and prosecution deadlines
- ☐ Review continuation/divisional strategy with counsel

Current USPTO Formatting / Submission Notes

For a nonprovisional utility application, the USPTO states that the specification, claims, and abstract should be filed in DOCX to avoid the non-DOCX surcharge, and that drawings may be submitted separately. The USPTO also describes the required application components, drawing standards, and oath/declaration requirements. Verify the current rules and fees in Patent Center immediately before filing.

INVENTION DISCLOSURE AND CLAIM STRATEGY

Internal drafting memorandum

Invention Statement

The proposed invention is an AI-controlled high-performance uncrewed aircraft architecture in which learned autonomy is not allowed to directly actuate the aircraft without deterministic constraint checking. The aircraft continuously combines sensor confidence, aircraft health, communication state, and authenticated human authority into the decision path.

Primary Claim Family

Core combination: heterogeneous sensor fusion + confidence/provenance state + AI candidate generation + deterministic safety-envelope gate + health-aware adaptation.

Secondary Claim Families

- Authenticated mission contracts and model lifecycle control.
- Degraded communications and local/remote state reconciliation.
- Forward simulation and shadow-mode evaluation of AI actions.
- Tamper-evident event logging and reconstruction.
- Fallback mode management based on confidence, health, and link state.

Potential Examiner Issues

- 35 U.S.C. 101: emphasize the concrete aircraft-control improvement and hardware interaction rather than an abstract decision process.
- 35 U.S.C. 102/103: expect combination arguments over known UAV planners, sensor fusion, and health monitoring. The specification should make the cooperation among these subsystems concrete.
- 35 U.S.C. 112(a): provide implementation detail, data structures, thresholds, examples, and alternative embodiments sufficient for enablement without tying the invention to one vendor or model architecture.
- 35 U.S.C. 112(b): keep terms such as “confidence,” “authorized action,” and “safety envelope” operationally defined in the specification.

Trade Secret / Patent Boundary

Before filing, decide which details must be disclosed to establish enablement and which implementation details are better preserved as trade secrets. Do not publish source code, model weights, test ranges, sensitive military operating parameters, or classified information in an unclassified patent filing unless appropriately authorized.

Claim Drafting Note

The claims intentionally avoid reciting weapon targeting, weapon release, or lethal engagement logic. The platform may be described as fighter-type in the specification only as an airframe class; the claimed improvements are in controlled autonomy and flight management.

AI-CONTROLLED HIGH-PERFORMANCE UNCREWED AIRCRAFT WITH CONFIDENCE-GATED ADAPTIVE FLIGHT MANAGEMENT

Provisional application draft - optional early filing

This provisional draft is intended to preserve an early filing date for the disclosed subject matter if filed properly and within applicable requirements. A provisional is not examined on the merits and should be followed by a nonprovisional within the permitted period if patent protection is pursued.

Inventor(s)

[INSERT EXACT LEGAL NAMES]

Disclosure

1. Aircraft Platform

The system comprises a high-performance fixed-wing uncrewed aircraft with propulsion, flight-control surfaces, power, communications, heterogeneous sensors, onboard computing, and storage.

2. AI Autonomy

An onboard AI planner generates candidate navigation and flight-management actions from a fused aircraft/environment state.

3. Confidence and Provenance

The state estimator attaches uncertainty/confidence and source provenance to state estimates and candidate actions.

4. Safety Gating

A deterministic safety-envelope engine independently evaluates candidate actions before they reach the aircraft controller.

5. Health-Aware Adaptation

A health estimator predicts component degradation and causes planning constraints and recovery margins to adapt before a failure becomes critical.

6. Human Supervision

Human instructions are encoded as cryptographically authenticated mission contracts with authorization scope and validity conditions.

7. Communications Resilience

The system detects link degradation, limits executable behaviors, preserves local state, and reconciles remote and local records after link recovery.

8. Evidence

The aircraft records model version, confidence, health state, command provenance, safety decisions, and communications transitions in a tamper-evident log.

9. Alternatives

Any of the above functions may be centralized or distributed across multiple processors or remote systems, provided the aircraft retains local enforcement of protected safety constraints.

Example Use Contexts

Training, flight research, lawful surveillance, search and rescue, communications relay, environmental monitoring, logistics, and other lawful missions requiring high-performance autonomous flight. No weapon targeting or engagement functionality is required by this disclosure.

README AND ATTORNEY REVIEW CHECKLIST

UCAS fighter-type AI aircraft patent package

What This Package Is

A structured patent drafting package intended to accelerate review by a registered patent attorney or patent agent. It includes a specification, claims, abstract, application-data worksheet, declaration worksheet, IDS draft, filing checklist, invention-disclosure strategy, provisional draft, and drawings.

What It Is Not

It is not a filed patent application, legal opinion, freedom-to-operate opinion, patentability opinion, or substitute for official USPTO forms and filing procedures.

Attorney Review Questions

- Who are the true inventors of each claimed feature?
- Has the invention been publicly disclosed, demonstrated, sold, offered for sale, or described to any third party?
- Was government funding used, and if so, does the application need a government-interest statement?
- Does a provisional or foreign application need to be claimed?
- Should the claims be split into multiple continuations or divisionals?
- Does the client want a broader or narrower claim strategy?
- Should sensitive implementation details remain proprietary instead of being disclosed in the application?

Recommended Filing Sequence

1. Confirm inventorship, ownership, priority, and disclosure history. 2. Review and revise the specification and claims. 3. Expand the prior-art search. 4. Update the IDS. 5. Populate the official ADS and declaration. 6. Review drawings against every claim element. 7. File through Patent Center and preserve the filing receipt and application number.

FIGURE 1 - High-Performance Uncrewed Aircraft

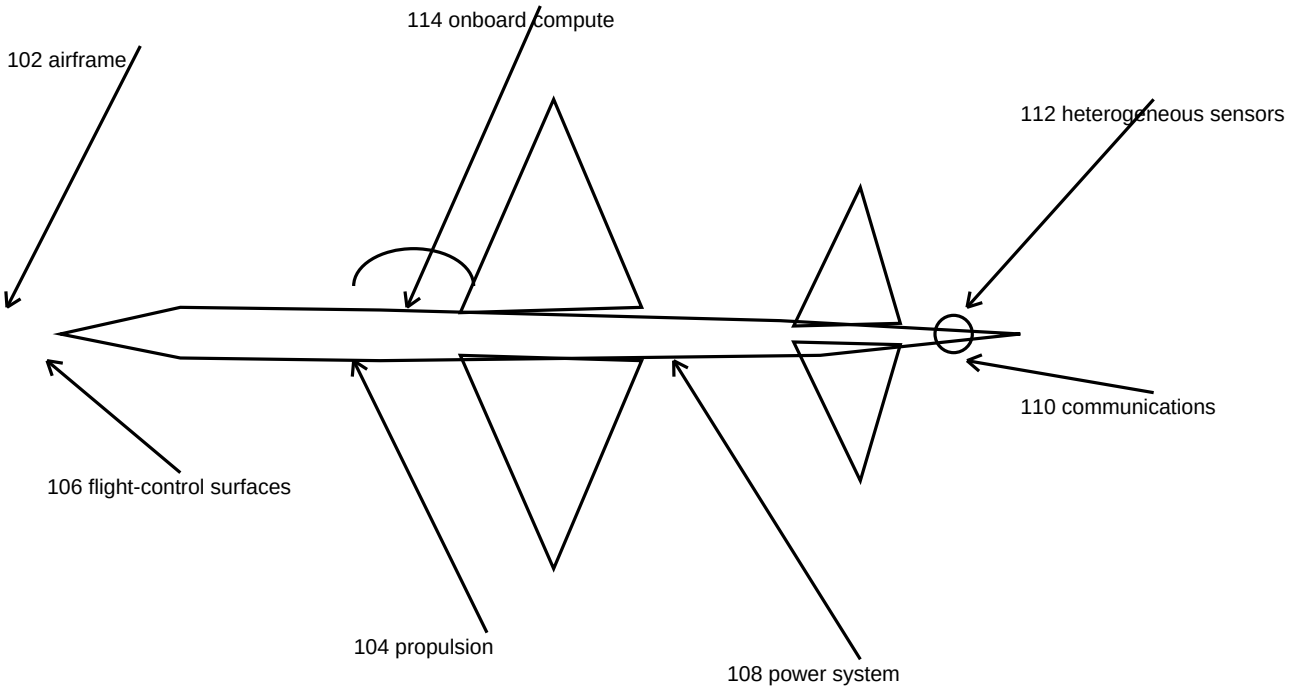


FIGURE 2 - Onboard Autonomy Stack

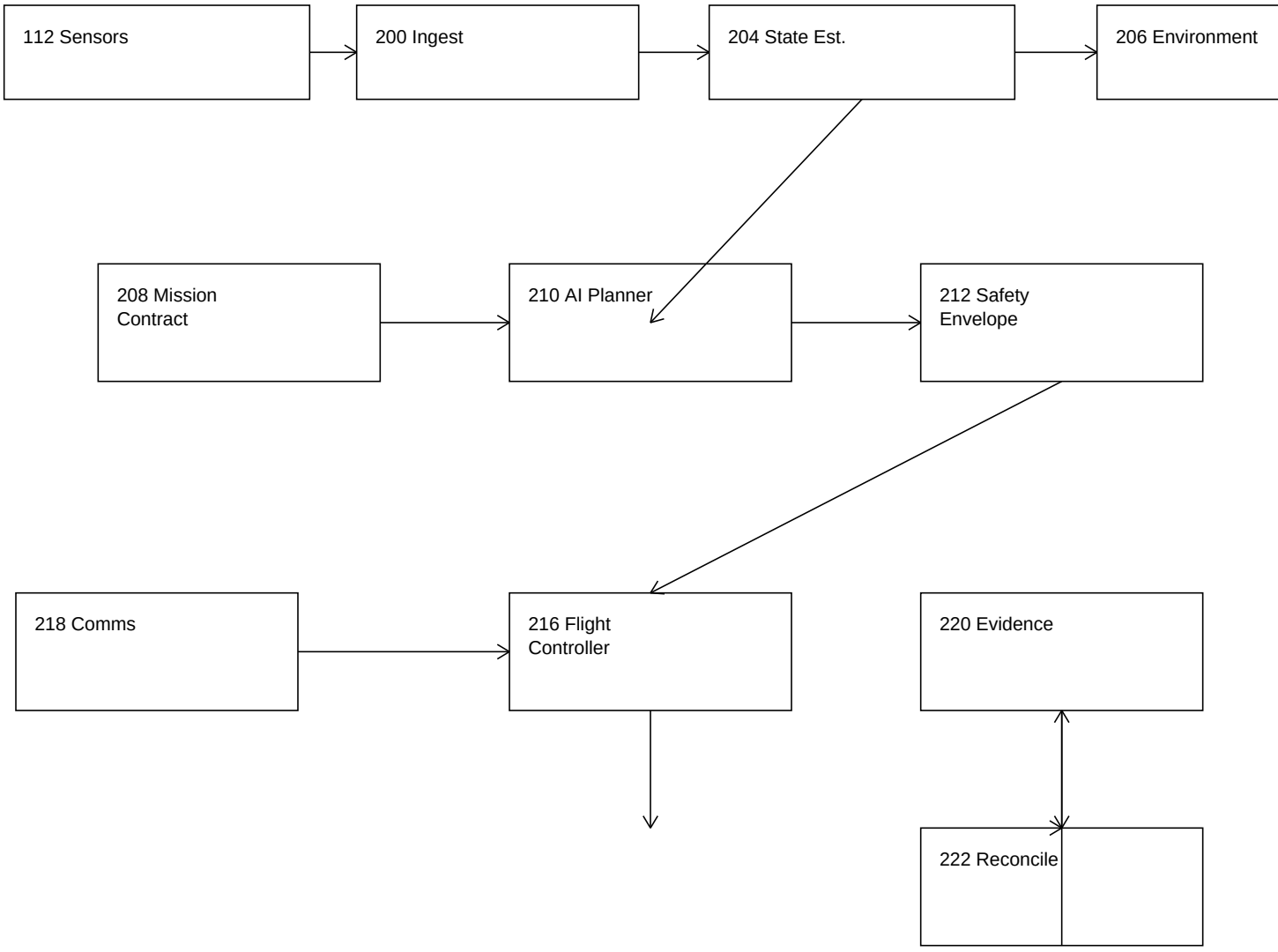


FIGURE 3 - Confidence-Aware Sensor Fusion

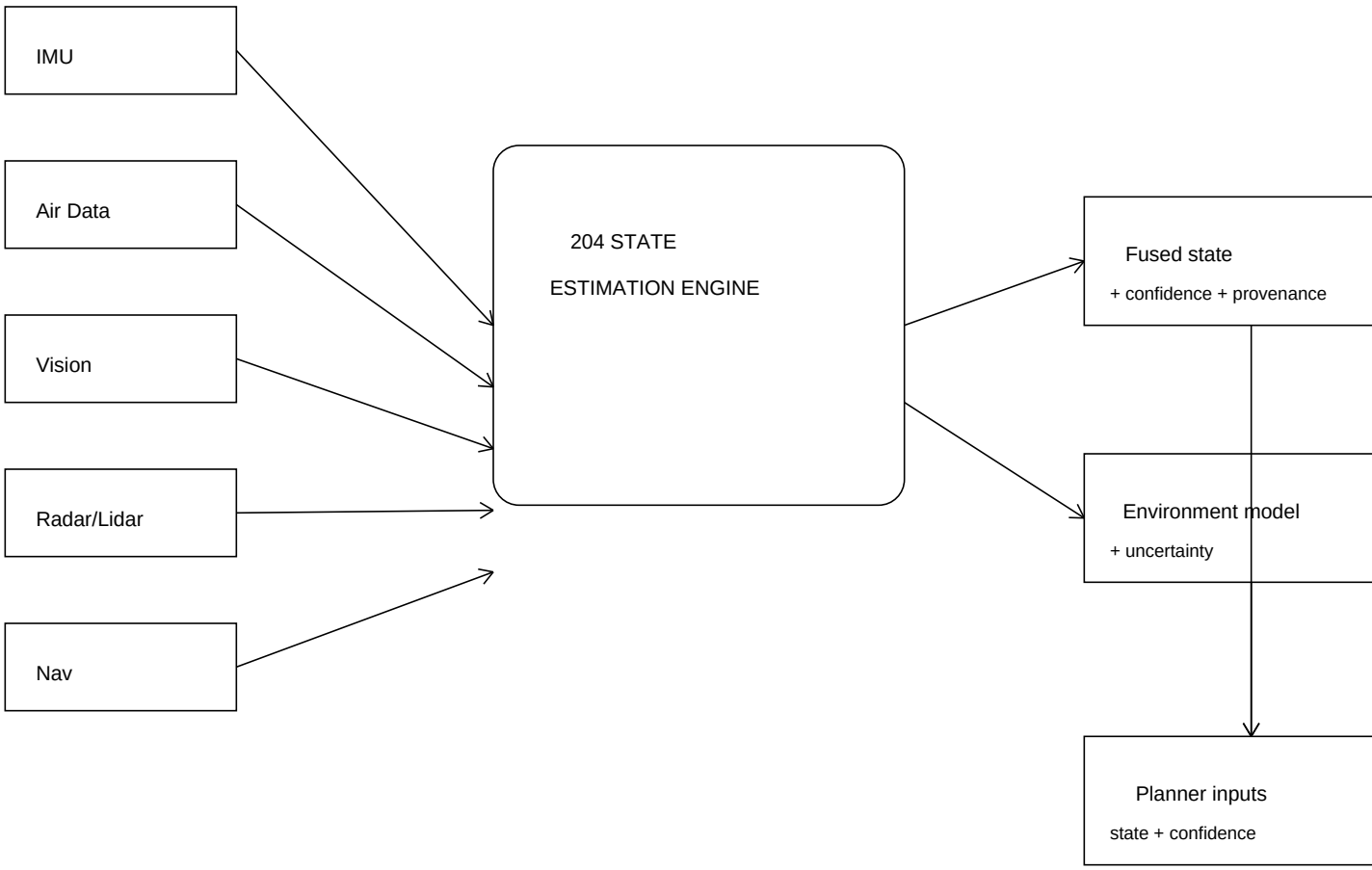


FIGURE 4 - AI Planning with Deterministic Safety Gating

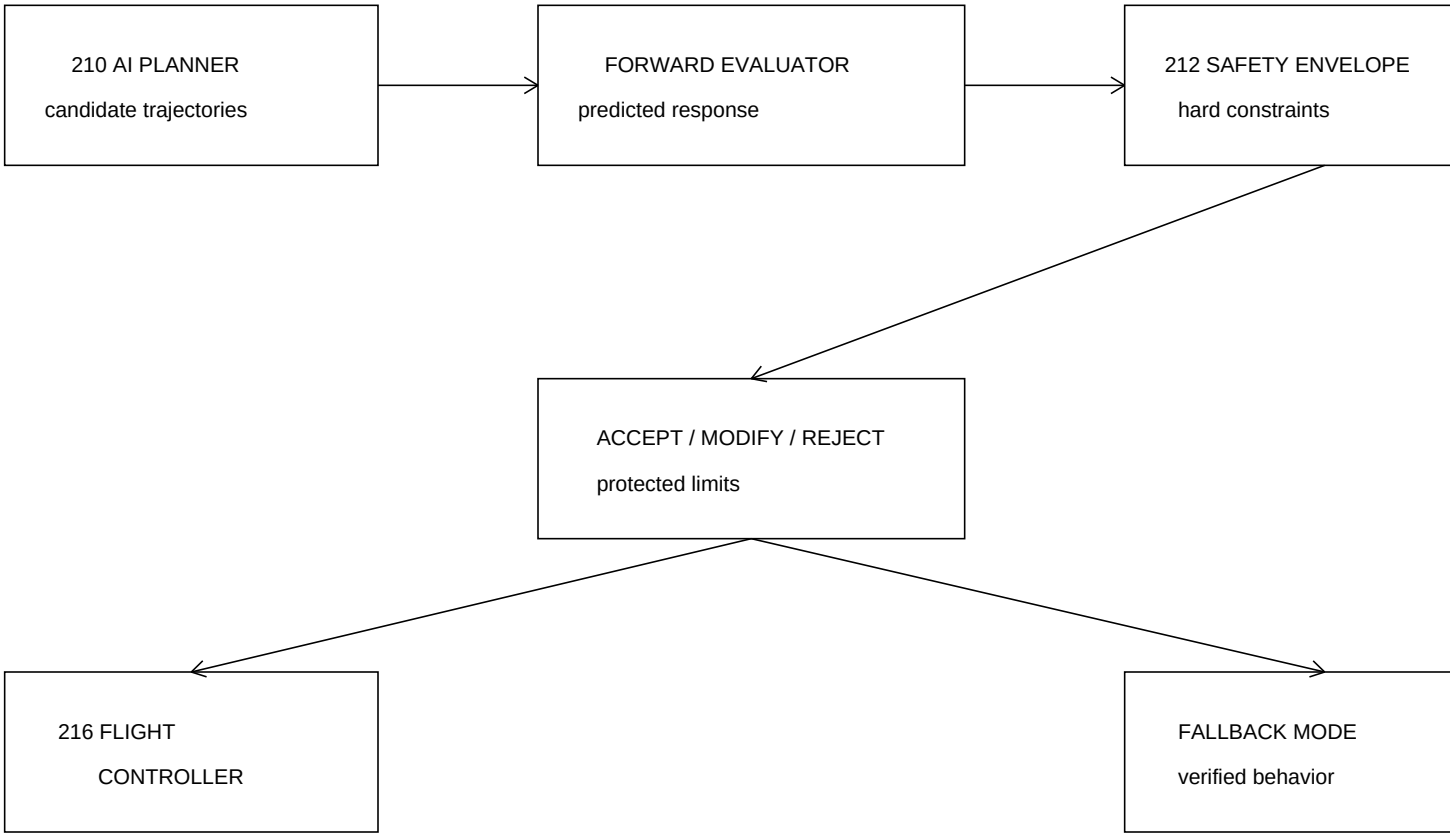
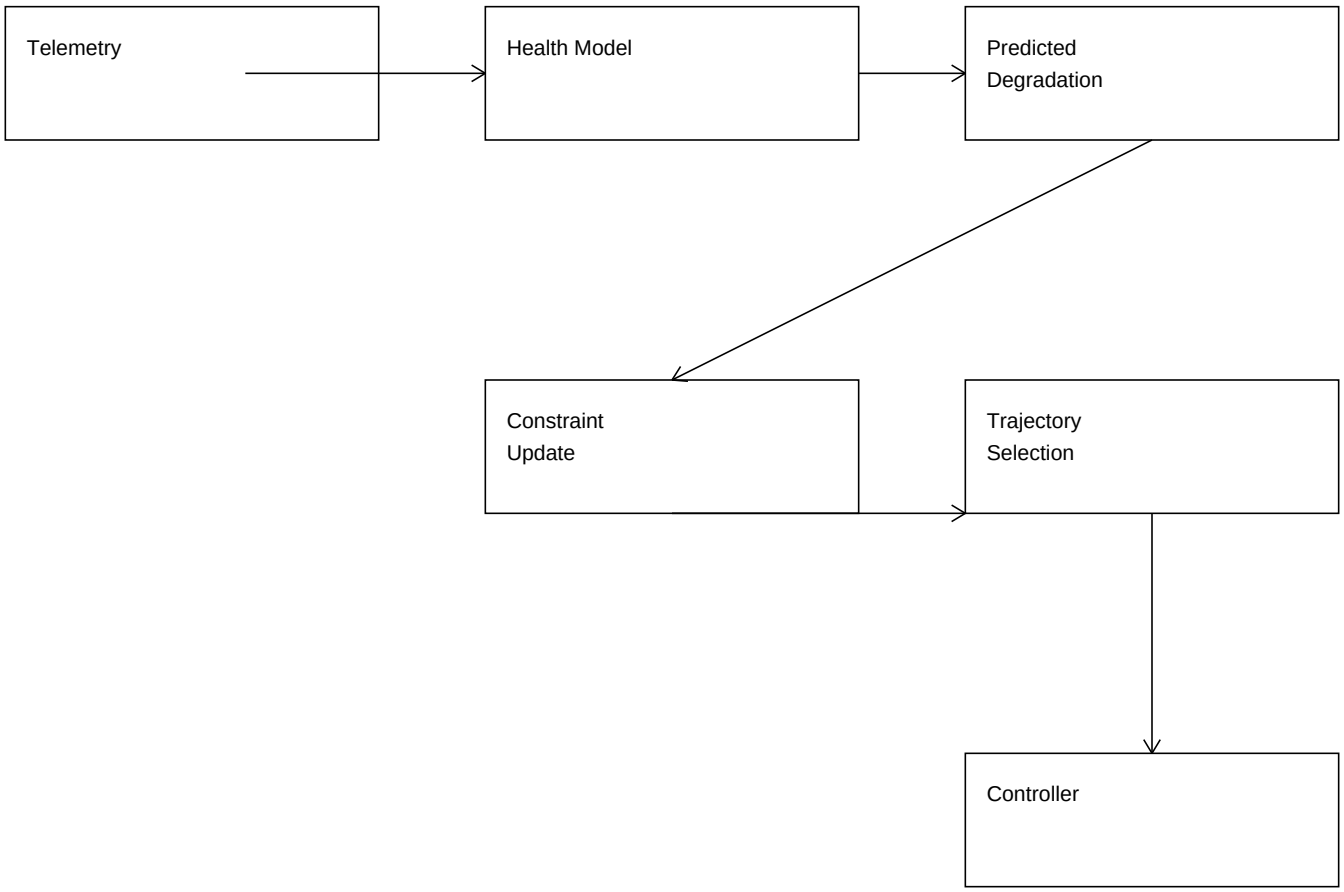


FIGURE 5 - Health-Aware Flight Reconfiguration



Health state changes planner constraints before a trajectory reaches the controller.

FIGURE 6 - Authenticated Human Supervision and Resilient Communications

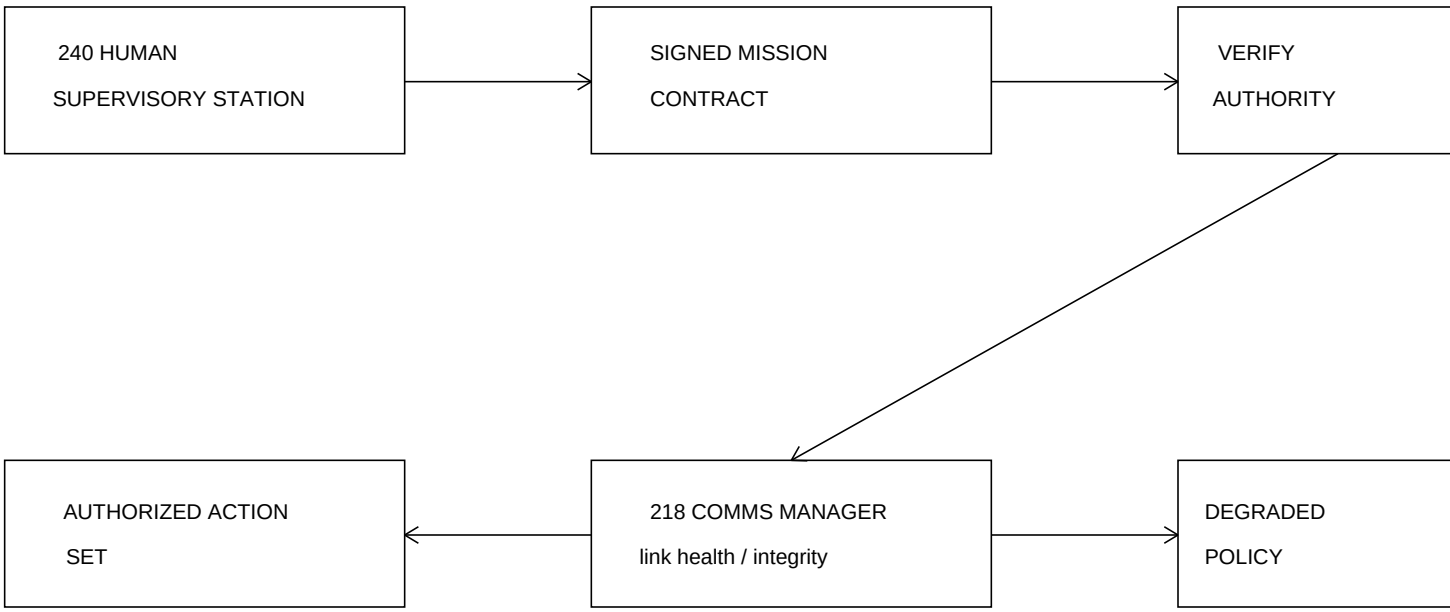
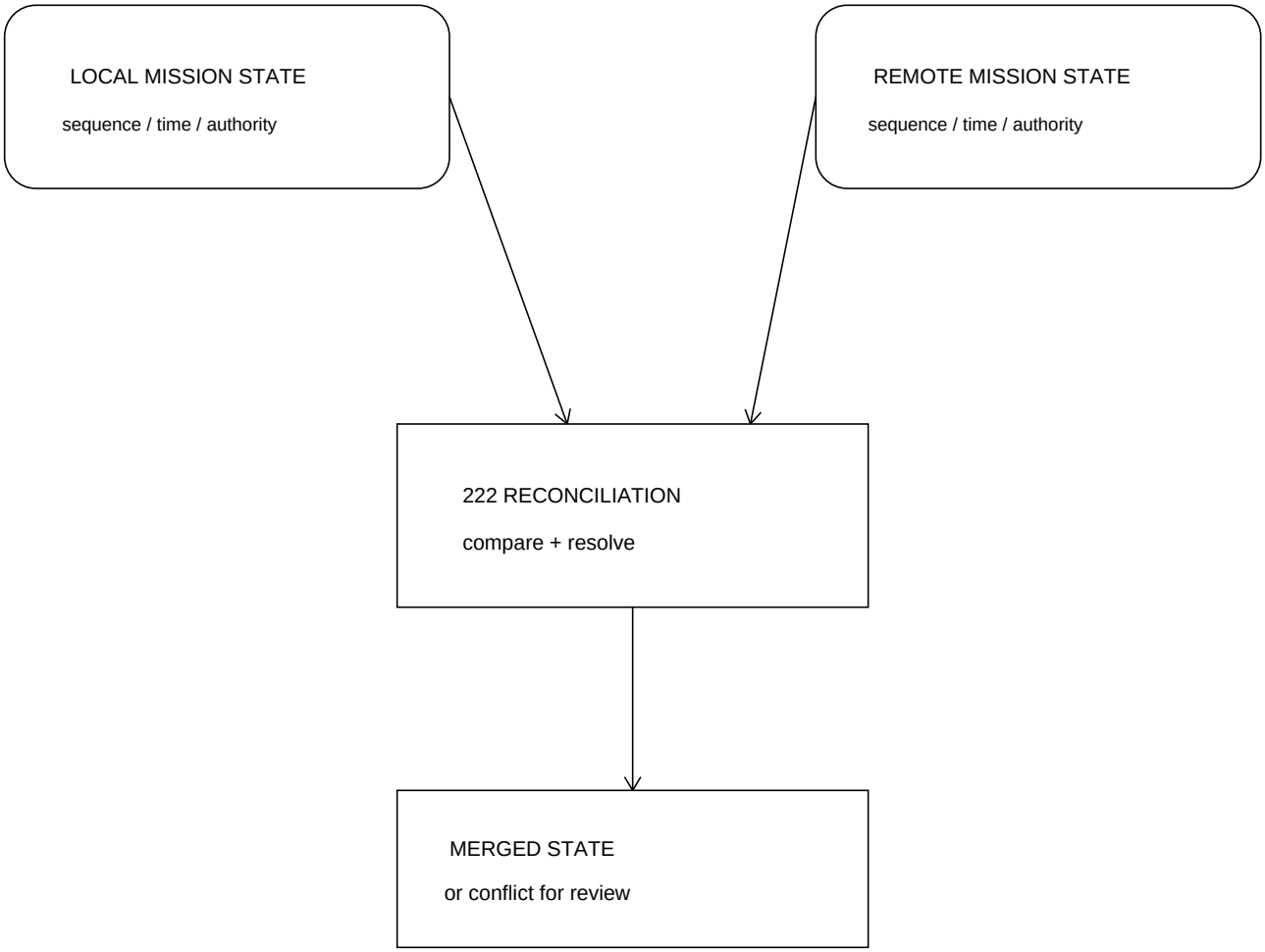
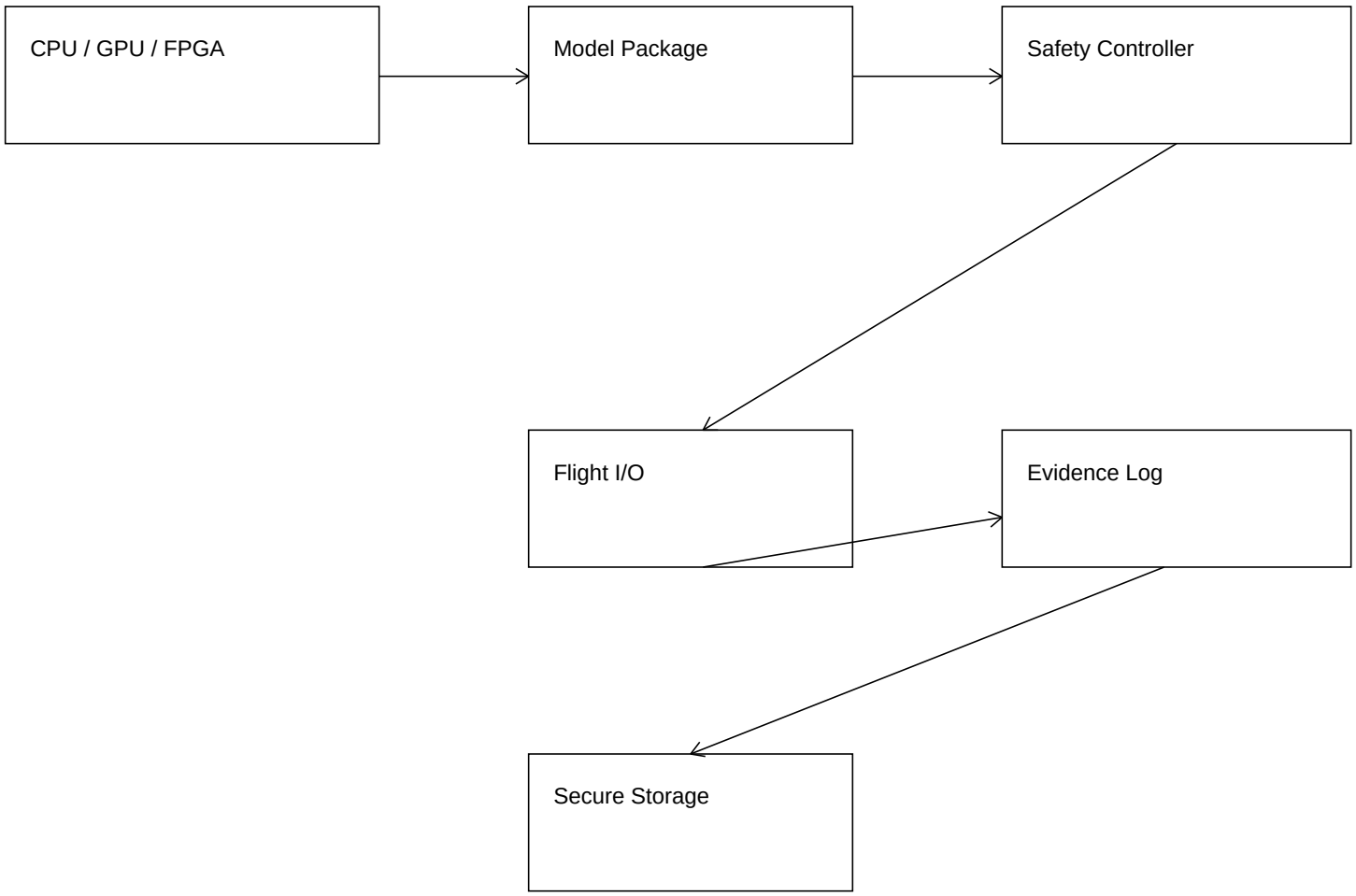


FIGURE 7 - Mission-State Reconciliation



Authority rules: revocation > stale command; unresolved conflicts remain non-executable.

FIGURE 8 - Onboard Compute and Evidence Logging



Evidence may include model identifier, state snapshot, confidence, health state, command provenance, and safety-envelope decision. Integrity may be protected by hash chaining or equivalent mechanisms.